

Srinjoy Sinha

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SUMMARY

AI & Full-Stack Engineer focused on production-grade Retrieval-Augmented Generation systems, secure enterprise workflow platforms, and scalable backend architectures. Experienced in multimodal data ingestion, vector search, transactional database design, and real-time LLM streaming with strong emphasis on security, performance, and clean system design.

CORE SKILLS

- **AI Systems & LLM Engineering:** Retrieval-Augmented Generation, FAISS, Ollama, Semantic Search & Embeddings, Tesseract, Prompt Engineering & Token Window Management, PyTorch, TensorFlow, scikit-learn
- **Backend & APIs:** Python (FastAPI), Node.js, Express.js, RESTful Architecture, Asynchronous APIs (asyncio, Server-Sent Events), Transactional SQL Design, MySQL Indexing & Pooling, Pydantic, Joi Validation, Swagger / OpenAPI
- **Frontend Engineering:** React 19, TypeScript, JavaScript (ES6+), Tailwind CSS 4, Vite, Context API, useReducer, Code Splitting (React.lazy, Suspense), WebGL (OGL), GLSL
- **Databases & Storage:** MySQL, PostgreSQL, MongoDB, FAISS Vector Store
- **Security & Systems:** JWT Authentication, RBAC, Bcrypt Hashing, Helmet, Rate Limiting, SQL Parameterization, CORS Policies
- **Dev Tools:** Git, Docker, ExcelJS, Multer, VS Code

EXPERIENCE

Jindal Power Limited

Jan 2026 - Mar 2026

AI & Full-Stack Intern

Designed and delivered enterprise-grade internal systems including:

- Jindal AI Operations Assistant
- Office Order Management System
- JPL Pass Approval System

PROJECTS

Jindal AI Operations Assistant (Production-Grade RAG System)

Jindal Power Limited

- Built a multimodal RAG pipeline enabling semantic search across 100+ enterprise documents (PDF, XLSX, PPTX, CSV).
- Implemented OCR fallback with validation heuristics to ensure high-quality embeddings.
- Developed FastAPI backend with Server-Sent Events (SSE) for real-time LLM streaming.
- Optimized local LLM orchestration using semaphore-based concurrency control.
- Designed session-aware prompt memory management within token constraints.

Office Order Management System (Enterprise HR Platform)

Jindal Power Limited

- Developed secure HR workflow system supporting 500+ employees with RBAC-based access control.
- Implemented atomic bulk uploads using Multer and MySQL transactions to maintain database–filesystem consistency.
- Built reusable transaction wrapper utilities for rollback safety across multi-step operations.
- Designed SQL-powered dashboard analytics for real-time HR insights.

JPL Pass Approval System (Workflow Automation Platform)

Jindal Power Limited

- Engineered full-stack gate-pass approval and material tracking system for 100+ users.
- Designed layered MVC architecture with centralized middleware-driven RBAC enforcement.
- Implemented transactional SQL logic for atomic multi-stage approval flows.
- Integrated Swagger documentation and Excel export reporting.

CERTIFICATIONS

- Artificial Intelligence & Machine Learning – Microsoft Learn × FICE
- Cloud Administration & Engineering – Microsoft Learn × FICE
- Microsoft Azure – Microsoft Learn × FICE
- Microsoft Copilot – Microsoft Learn × FICE
- Power BI – Microsoft Learn × FICE
- Deloitte Technology & Data Analytics Job Simulation

EDUCATION

Heritage Institute of Technology, Kolkata

Bachelor of Technology, Computer Science Engineering

- GPA: 7.53 / 10

2023 - 2027

Kolkata